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Problem Set 1 Solutions

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Dynamic Programming: An Analytical Case

Question 1: Consider the standard growth model with a logarithmic utility function and full depreciation. The social planning problem is:

$$\max_{\{c_t,k_{t+1}\}_{t=0}^{\infty}} \left\{ \sum_{t=0}^{\infty} \beta^t \log(c_t) \right\}$$

subject to:

$$c_t + k_{t+1} = k_t^{\alpha},$$

with $k_0 > 0$ given.

1. Formulate a Bellman equation for the social planning problem.

   Answer: The Bellman equation is:

   $$v(k) = \max_{k'} \left\{ \log(k^{\alpha} - k') + \beta v(k') \right\}$$

2. It turns out that the Bellman equation can be solved analytically. Conjecture that the value function takes the form:

   $$v(k) = a_0 + a_1 \log(k).$$

   Solve for the coefficient a_1 and for the optimal policy function $k' = g(k)$, and find the coefficient a_0. Hint: after plugging in the conjectured functional form, you can take first-order and envelope conditions and solve the resulting system of equations.

   Answer: Plugging in the guess gives:

   $$a_0 + a_1 \log(k) = \max_{k'} \left\{ \log(k^{\alpha} - k') + \beta(a_0 + a_1 \log(k')) \right\} \tag{1}$$

   The first-order condition for k' is:

   $$-\frac{1}{k^{\alpha} - k'} + \beta \frac{a_1}{k'} = 0$$

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or:

$$\beta \frac{a_1}{k'} = \frac{1}{k^\alpha - k'}. \tag{2}$$

The envelope condition for k (i.e. taking a derivative with respect to k on both sides) is:

$$\frac{a_1}{k} = \frac{\alpha k^{\alpha-1}}{k^\alpha - k'}.$$

Solving this for a_1 and plugging the result into (2) gives:

$$\beta \frac{\alpha k^\alpha}{k'(k^\alpha - k')} = \frac{1}{k^\alpha - k'}.$$

Solving for k' yields the policy function:

$$k' = \alpha\beta k^\alpha.$$

Plugging this back into (2) and solving for a_1 gives:

$$a_1 = \frac{\alpha}{1 - \alpha\beta}.$$

To solve for a_0, we can plug the solution for k' into (1):

$$a_0 + a_1 \log(k) = \log((1 - \alpha\beta)k^\alpha) + \beta(a_0 + a_1 \log(\alpha\beta k^\alpha))$$

The equation has to hold for all k, including $k = 1$. Plugging in $k = 1$ gives:

$$a_0 = \log(1 - \alpha\beta) + \beta(a_0 + a_1 \log(\alpha\beta)).$$

We therefore have:

$$a_0 = \frac{1}{1 - \beta} \left(\log(1 - \alpha\beta) + \beta a_1 \log(\alpha\beta)\right).$$

Additional remarks

How to guess the functional form. The guess $v(k) = a_0 + a_1 \log(k)$ can be found by iterating on the Bellman equation, starting from $v_0(k) = 0$. With $v_0(k) = 0$, the future has no value:

$$v_1(k) = \max_{k'} \left\{\log(k^\alpha - k')\right\},$$

so that $k' = 0$ and $v_1(k) = \alpha \log(k)$. Using v_1 as the continuation value gives:

$$v_2(k) = \max_{k'} \left\{\log(k^\alpha - k') + \alpha\beta \log(k')\right\}.$$

The first-order condition $1/(k^\alpha - k') = \alpha\beta/k'$ gives $k' = (\alpha\beta/(1+\alpha\beta))k^\alpha$, and substituting back:

$$v_2(k) = \alpha\beta\log(\alpha\beta) - (1+\alpha\beta)\log(1+\alpha\beta) + \alpha(1+\alpha\beta)\log(k).$$

Every iteration has the form $v_n(k) = a_{0,n} + a_{1,n}\log(k)$, with

$$a_{1,n} = \alpha\left(1 + \alpha\beta + \ldots + (\alpha\beta)^{n-1}\right),$$

which converges to $a_1 = \alpha/(1-\alpha\beta)$. Iteration n is the problem of a household with n periods left, with saving rate

$$\alpha\beta\,\frac{1-(\alpha\beta)^{n-1}}{1-(\alpha\beta)^n},$$

which goes from 0 (last period) to $\alpha\beta/(1+\alpha\beta)$ (two periods left) and converges to $\alpha\beta$ as $n \to \infty$.

Other cases with a closed-form value function. Closed-form solutions are the exception. The main cases are:

1. Logarithmic utility, Cobb-Douglas production, and full depreciation (this question). The solution extends to productivity shocks $\log z' = \rho\log z + \varepsilon'$: the value function becomes $v(k,z) = a_0 + a_1\log(k) + a_2\log(z)$, with the same a_1, and $k' = \alpha\beta z k^\alpha$.
2. The same case with valued leisure l: with utility $\log(c) + v(l)$, where v is increasing and concave, and output $k^\alpha(1-l)^{1-\alpha}$, the saving rate is still $\alpha\beta$ and leisure l is constant.
3. CRRA utility and linear production, for any δ (the AK model of Question 3).
4. Quadratic utility and linear production: the value function is quadratic and the policy function is linear.

In the standard model of Question 2 with $\delta < 1$ or $\sigma \neq 1$, there is no closed-form solution, and the value function has to be computed numerically.

The Standard Model

Question 2: Consider the standard optimal growth model represented by the planning problem:

$$\max_{\{c_t, k_{t+1}\}_{t=0}^\infty} \left\{ \sum_{t=0}^\infty \beta^t \frac{c_t^{1-\sigma}}{1-\sigma} \right\}.$$

subject to:

$$\begin{aligned} c_t + k_{t+1} &\leq (1-\delta)k_t + A_t(k_t)^{\alpha}(n_t)^{1-\alpha}, \\ A_{t+1} &= \gamma A_t, \\ n_t &= 1, \end{aligned}$$

with $\gamma > 1$, $\sigma > 0$, $\alpha, \beta, \delta \in (0,1)$, and A_0, k_0 given.

1. Consider a market structure in which a representative firm buys labor from households at wage w_t and rents capital at interest rate r_t. Define a competitive equilibrium for this economy.

   Answer: An equilibrium consists of an allocation sequence $\{c_t, k_{t+1}, n_t\}_{t=0}^{\infty}$ and a sequence of prices $\{r_t, w_t\}_{t=0}^{\infty}$ such that:

   - Given prices, the allocation solves the consumer utility maximization problem:

     $$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \left\{ \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma}}{1-\sigma} \right\}$$

     subject to:

     $$\begin{aligned} c_t + k_{t+1} &\leq (1-\delta+r_t)k_t + w_t n_t, \\ n_t &= 1, \\ k_{t+1} &\geq 0. \end{aligned}$$

   - Given prices, the choices for capital and labor in period t solve:

     $$\max_{k_t, n_t} \left\{ A_t(k_t)^{\alpha}(n_t)^{1-\alpha} - r_t k_t - w_t n_t \right\}$$

   - The allocation is feasible, i.e., at each date t we have:

     $$c_t + k_{t+1} = (1-\delta)k_t + A_t(k_t)^{\alpha}(n_t)^{1-\alpha}.$$

2. The equilibrium turns out to be characterized by convergence to a balanced growth path. To characterize this balanced growth path, it turns out to be useful to rewrite the model in terms of variables that are detrended by productivity growth. For example, the detrended consumption $\tilde{c}_t$ is given by:

   $$\tilde{c}_t = \frac{c_t}{\tilde{A}_t},$$

   where $\tilde{A}_t = A_t^{1/(1-\alpha)}$. Formulate a Bellman equation for the social planning problem for this economy in terms of the detrended variables.

Answer: Define $\tilde{A}_t = A_t^{1/(1-\alpha)} = A_0^{1/(1-\alpha)} \left(\gamma^{1/(1-\alpha)}\right)^t$ and define $\tilde{c}_t = c_t/\tilde{A}_t$ and $\tilde{k}_t = k_t/\tilde{A}_t$. Solving the definition for A_t we get:

$$A_t = \tilde{A}_t^{1-\alpha}.$$

Plugging this into the production function $A_t(k_t)^\alpha(n_t)^{1-\alpha}$ we can write the production function as $(k_t)^\alpha(\tilde{A}_t n_t)^{1-\alpha}$. Solving the other definitions we get:

$$\begin{aligned} c_t =& \tilde{c}_t \tilde{A}_t, \\ k_t =& \tilde{k}_t \tilde{A}_t. \end{aligned}$$

Plugging in these expressions, the planning problem can then be written as:

$$\max_{\{\tilde{c}_t, \tilde{k}_{t+1}\}_{t=0}^\infty} \left\{ \sum_{t=0}^\infty \beta^t \frac{\tilde{c}_t^{1-\sigma} \tilde{A}_t^{1-\sigma}}{1-\sigma} \right\}.$$

subject to:

$$\begin{aligned} \tilde{c}_t \tilde{A}_t + \tilde{k}_{t+1} \tilde{A}_{t+1} &\leq (1-\delta)\tilde{k}_t \tilde{A}_t + (\tilde{k}_t \tilde{A}_t)^\alpha (\tilde{A}_t n_t)^{1-\alpha}, \\ \tilde{A}_{t+1} &= \gamma^{\frac{1}{1-\alpha}} \tilde{A}_t, \\ n_t &= 1. \end{aligned}$$

We can divide by $\tilde{A}_t$ in the resource constraint and use:

$$\tilde{A}_t = \tilde{A}_0 \left(\gamma^{\frac{1}{1-\alpha}}\right)^t$$

in the objective to write the planning problem as:

$$\max_{\{\tilde{c}_t, \tilde{k}_{t+1}\}_{t=0}^\infty} \left\{ \sum_{t=0}^\infty \left(\beta \gamma^{\frac{1-\sigma}{1-\alpha}}\right)^t \frac{\tilde{c}_t^{1-\sigma} \tilde{A}_0^{1-\sigma}}{1-\sigma} \right\}.$$

subject to:

$$\begin{aligned} \tilde{c}_t + \tilde{k}_{t+1} \gamma^{\frac{1}{1-\alpha}} &\leq (1-\delta)\tilde{k}_t + (\tilde{k}_t)^\alpha (n_t)^{1-\alpha}, \\ n_t &= 1. \end{aligned}$$

This is now a stationary problem with the corresponding Bellman equation (omitting the constant $\tilde{A}_0$ that does not affect decisions):

$$v(\tilde{k}) = \max_{\tilde{k}'} \left\{ \frac{\tilde{c}^{1-\sigma}}{1-\sigma} + \tilde{\beta} v(\tilde{k}') \right\}$$

s.t.

$$\tilde{c} + \tilde{k}' \gamma^{\frac{1}{1-\alpha}} \leq (1-\delta)\tilde{k} + \tilde{k}^{\alpha}$$

where $\tilde{\beta} = \beta\gamma^{(1-\sigma)/(1-\alpha)}$. In practice, if $\sigma > 1$, $\tilde{\beta} < \beta$ and in the solution it is as if the household were more impatient in the transformed problem. The opposite when $\sigma < 1$. We assume $\tilde{\beta} < 1$, so that the problem is well defined (utility is bounded along the balanced growth path): notice that this requirement collapses to $\beta < 1$ when $\sigma > 1$, since in this case CRRA utility is bounded from above. However, the requirement becomes more restrictive when $\sigma < 1$, since we need $\beta < \gamma^{(\sigma-1)/(1-\alpha)}$. In practice we can impose $\beta < \min\{\gamma^{(\sigma-1)/(1-\alpha)}, 1\}$.

Note the factor $\gamma^{1/(1-\alpha)}$ multiplying $\tilde{k}'$ in the detrended resource constraint. It appears because next period's capital is divided by today's $\tilde{A}_t$ rather than by $\tilde{A}_{t+1}$: $k_{t+1}/\tilde{A}_t = \gamma^{1/(1-\alpha)}\tilde{k}_{t+1}$. Since effective labor grows at rate $\gamma^{1/(1-\alpha)}$, keeping capital per unit of effective labor constant requires extra saving today. Technology growth therefore dilutes detrended capital, just like population growth in the Solow model. On the balanced growth path, detrended investment has to cover both depreciation and growth:

$$\tilde{\imath} = \tilde{k}^{\alpha} - \tilde{c} = \left(\gamma^{\frac{1}{1-\alpha}} - 1 + \delta\right)\tilde{k}.$$

3. Use first-order and envelope conditions to characterize the balanced growth path (i.e., the path in which output, consumption, and capital all grow at a common rate).

   Answer: The first-order and envelope conditions are:

$$\frac{\gamma^{\frac{1}{1-\alpha}}}{\tilde{c}^{\sigma}} = \tilde{\beta} v'(\tilde{k}'),$$
$$v'(\tilde{k}) = \frac{1-\delta+\alpha\tilde{k}^{\alpha-1}}{\tilde{c}^{\sigma}}.$$

   Combining these equations gives a version of the consumption Euler equation:

$$\frac{\gamma^{\frac{1}{1-\alpha}}}{\tilde{c}^{\sigma}} = \tilde{\beta}\frac{1-\delta+\alpha\tilde{k}'^{\alpha-1}}{(\tilde{c}')^{\sigma}}.$$

   In the balanced growth path we have $\tilde{c} = \tilde{c}'$ and hence:

$$\gamma^{\frac{1}{1-\alpha}} = \tilde{\beta}(1-\delta+\alpha\tilde{k}'^{\alpha-1}),$$

   which can be solved for the constant $\tilde{k}$ in the balanced growth path:

$$\tilde{k} = \left(\frac{\alpha}{\gamma^{\frac{1}{1-\alpha}}/\tilde{\beta} + \delta - 1}\right)^{\frac{1}{1-\alpha}}.$$

For the balanced growth facts, the important fact is that $\tilde{k}$ is constant in the balanced growth path.

4. Assuming that interest rates and wages are given by the marginal products of capital and labor, demonstrate that in the balanced growth path the economy satisfies the following observations on long-run growth:

   (a) The income shares of capital and labor are constant.
   Answer: The labor income share is:

$$\begin{aligned}\frac{w_t n_t}{y_t} &= \frac{w_t n_t}{A_t (k_t)^{\alpha} (n_t)^{1-\alpha}} \\ &= \frac{(1-\alpha) A_t (k_t)^{\alpha} (n_t)^{1-\alpha}}{A_t (k_t)^{\alpha} (n_t)^{1-\alpha}} \\ &= 1-\alpha,\end{aligned}$$

   which is constant. Similarly, the capital share is:

$$\begin{aligned}\frac{r_t k_t}{y_t} &= \frac{r_t k_t}{A_t (k_t)^{\alpha} (n_t)^{1-\alpha}} \\ &= \frac{\alpha A_t (k_t)^{\alpha} (n_t)^{1-\alpha}}{A_t (k_t)^{\alpha} (n_t)^{1-\alpha}} \\ &= \alpha.\end{aligned}$$

   (b) The return on capital is constant.
   Answer: The return on capital is:

$$r_t = \alpha \left(\frac{\tilde{A}_t n_t}{k_t} \right)^{1-\alpha} = \alpha \left(\frac{1}{\tilde{k}_t} \right)^{1-\alpha}.$$

   The return on capital is thus solely a function of $\tilde{k}_t$, and given that $\tilde{k}$ is constant in the balanced growth path, the return on capital is constant also.

   (c) The capital-output ratio is constant.
   Answer: The capital output ratio is:

$$\frac{k_t}{y_t} = \left(\frac{k_t}{\tilde{A}_t n_t} \right)^{1-\alpha} = \left(\tilde{k}_t \right)^{1-\alpha}.$$

   The capital-output ratio is thus solely a function of $\tilde{k}_t$, and given that $\tilde{k}$ is constant in the balanced growth path, the capital output ratio is constant also.

(d) The wage increases at a constant rate.

Answer: The wage is given by:

$$w_t = (1-\alpha)\tilde{A}_t^{1-\alpha}\left(\frac{k_t}{n_t}\right)^{\alpha} = (1-\alpha)\tilde{A}_t\left(\frac{k_t}{\tilde{A}_t}\right)^{\alpha} = (1-\alpha)\tilde{A}_t\left(\tilde{k}_t\right)^{\alpha}.$$

Given that $\tilde{k}$ is constant in steady state, the wage rate in the balanced growth path is proportional to $\tilde{A}_t$, and thus grows at rate $\gamma^{1/(1-\alpha)}$.

5. Assume the economy is displaced from the balanced growth path because of a partial destruction of the capital stock (say, because of an earthquake). Which of the growth facts will continue to hold on the transition path back to the balanced growth path?

   Answer: The variable $\tilde{k}_t$ changes during the transition path. Thus, the only growth facts that continue to hold are constant factor income shares (which were not conditional on $\tilde{k}$ being constant, whereas the other statistics will change during the transition path).

   Figure 1 illustrates the transition after an earthquake that destroys 30 percent of the capital stock. On impact, the capital-output ratio falls and the return on capital rises, while the wage falls because each worker has less capital to work with. During the transition, the return on capital falls back, the capital-output ratio rises, and the wage grows faster than its trend rate $\gamma^{1/(1-\alpha)}$, until the economy is back on the original balanced growth path.

Additional remarks

This question studies the neoclassical growth model, known as the Ramsey–Cass–Koopmans model after Ramsey (1928), Cass (1965), and Koopmans (1965). It is the Solow model with an optimally chosen saving rate, and the fourth part of the question shows that its balanced growth path is consistent with Kaldor's (1961) growth facts. Nicholas Kaldor graduated from the LSE in 1930 and taught here from 1932 to 1947, before moving to Cambridge. As of 9 October 2026, the Nicholas Kaldor Professor of Economics at the LSE is Matthias Doepke (who is teaching this course!).

The AK Model

Question 3: Consider a growth model characterized by the planning problem:

$$\max_{\{c_t,k_{t+1}\}_{t=0}^{\infty}} \left\{\sum_{t=0}^{\infty}\beta^t\frac{c_t^{1-\sigma}}{1-\sigma}\right\}.$$

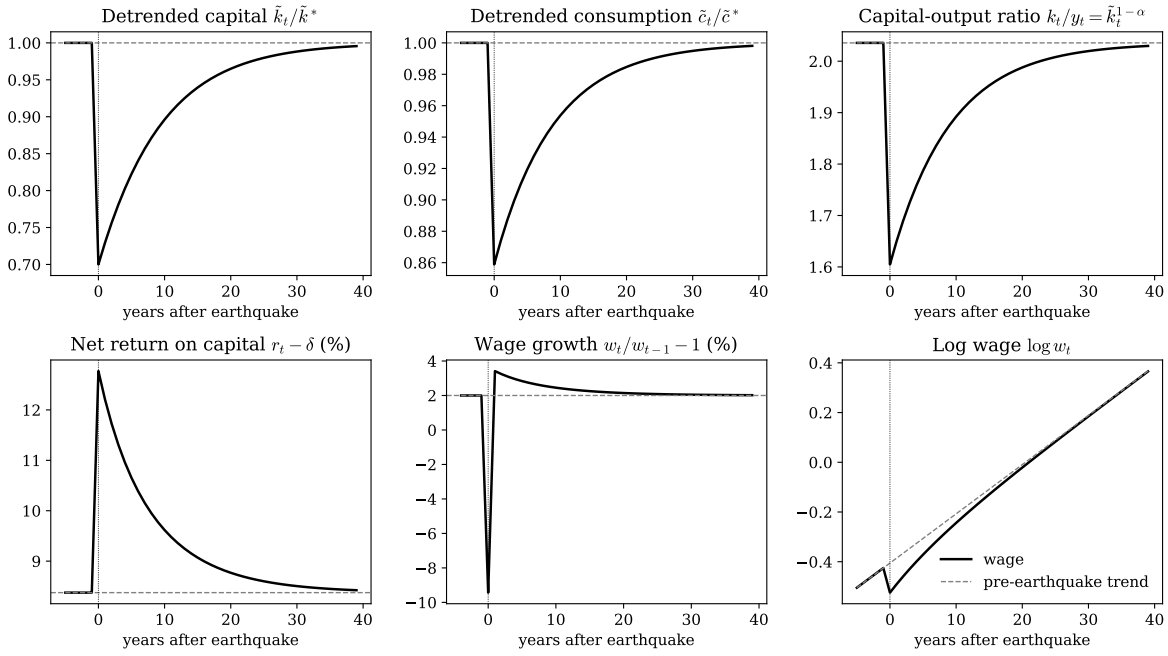


Figure 1: Transition after an earthquake that destroys 30 percent of the capital stock in period 0. Parameters: $\alpha = 1/3$, $\beta = 0.96$, $\delta = 0.08$, $\sigma = 2$, and $\gamma^{1/(1-\alpha)} = 1.02$ (one period is a year). Dashed lines show the balanced growth path.

subject to:

$$c_t + k_{t+1} \leq (1-\delta)k_t + Ak_t.$$

Assume $\sigma \neq 1$.

1. Make a direct argument showing that the value function $v^\star(k)$ for the sequential optimization problem should be of the form:

$$v^\star(k) = a\frac{k^{1-\sigma}}{1-\sigma}.$$

Answer: First, the value function evaluated at $k = 1$ is

$$v^\star(1) = \max_{\{k_{t+1}\}_{t=0}^{\infty}} \sum \beta^t u((1-\delta+A)k_t - k_{t+1})$$

s.t. $k_0 = 1$.

Let's now take some arbitrary other capital value $k > 0$. We have

$$v^\star(k) = \max_{\{k_{t+1}\}_{t=0}^{\infty}} \sum \beta^t u((1-\delta+A)k_t - k_{t+1})$$

s.t. $k_0 = k$.

We now do a change of variable and define $\tilde{k}_t = k_t/k$, so that $k_t = \tilde{k}_t k$. This is just a rescaling of variables, and does not change anything about the solution of the problem. Plugging in these rescaled variables, we get (the maximization is now in terms of $\tilde{k}_t$):

$$v^\star(k) = \max_{\{\tilde{k}_t\}} \sum \beta^t u((1-\delta+A)\tilde{k}_t k - \tilde{k}_{t+1} k)$$

s.t. $\tilde{k}_0 k = k$. Simplifying this expression, we get (given CRRA):

$$v^\star(k) = k^{1-\sigma} \max_{\{\tilde{k}_t\}} \sum \beta^t u((1-\delta+A)\tilde{k}_t - \tilde{k}_{t+1})$$

s.t. $\tilde{k}_0 = 1$, so that:

$$v^\star(k) = k^{1-\sigma} v^\star(1)$$

which has the required form. The value at $k = 1$ plays the role of the constant: matching $v^\star(k) = k^{1-\sigma}v^\star(1)$ with $v^\star(k) = ak^{1-\sigma}/(1-\sigma)$ gives $a = (1-\sigma)v^\star(1)$. Note that $a > 0$ for any σ, since $v^\star(1)$ has the same sign as $1-\sigma$.

2. Use the Bellman equation corresponding to the planning problem to compute a.

   Answer: The Bellman equation is:

$$v(k) = \max_{k'} \left\{ \frac{((1-\delta+A)k - k')^{1-\sigma}}{1-\sigma} + \beta v(k') \right\}$$

   By the principle of optimality, its solution v is the value function $v^\star$ of the sequential problem, so we guess $v(k) = ak^{1-\sigma}/(1-\sigma)$. Plugging in the guess gives:

$$a\frac{k^{1-\sigma}}{1-\sigma} = \max_{k'} \left\{ \frac{((1-\delta+A)k - k')^{1-\sigma}}{1-\sigma} + \beta a \frac{(k')^{1-\sigma}}{1-\sigma} \right\}$$

   The first order and envelope conditions are:

$$\begin{aligned} ((1-\delta+A)k - k')^{-\sigma} &= \beta a (k')^{-\sigma} \\ ak^{-\sigma} &= (1-\delta+A)((1-\delta+A)k - k')^{-\sigma} \end{aligned}$$

   Using the first equation to replace the last term on the right hand side of the second equation gives an expression that can be solved for the policy function:

$$\begin{aligned} ak^{-\sigma} &= (1-\delta+A)\beta a (k')^{-\sigma} \\ k' &= (\beta(1-\delta+A))^{\frac{1}{\sigma}} k. \end{aligned}$$

   Substituting this back into the first-order condition gives:

$$\begin{aligned} ((1-\delta+A)k - (\beta(1-\delta+A))^{\frac{1}{\sigma}}k)^{-\sigma} &= \beta a((\beta(1-\delta+A))^{\frac{1}{\sigma}}k)^{-\sigma} \\ ((1-\delta+A) - (\beta(1-\delta+A))^{\frac{1}{\sigma}})^{-\sigma} &= a((1-\delta+A)^{\frac{1}{\sigma}})^{-\sigma} \\ a &= \left(\frac{1-\delta+A}{(1-\delta+A)^{\frac{1}{\sigma}}} - \beta^{\frac{1}{\sigma}} \right)^{-\sigma} \\ a &= \left((1-\delta+A)^{\frac{\sigma-1}{\sigma}} - \beta^{\frac{1}{\sigma}} \right)^{-\sigma}. \end{aligned}$$

   For this solution to be valid, consumption

$$c = \left((1-\delta+A) - (\beta(1-\delta+A))^{\frac{1}{\sigma}} \right) k$$

   has to be positive, which requires $(1-\delta+A)^{(\sigma-1)/\sigma} > \beta^{1/\sigma}$, or equivalently $\beta(1-\delta+A)^{1-\sigma} < 1$. This condition also ensures that lifetime utility is finite.

3. Compute the equilibrium growth rate of the economy.

Answer: As output is proportional to capital, the economy grows at the same rate as the capital stock. Hence, the growth rate is:

$$\frac{k'}{k} - 1 = (\beta(1 - \delta + A))^{\frac{1}{\sigma}} - 1.$$

Consumption grows at the same rate, since it is a constant fraction of k. The growth rate is positive if $\beta(1 - \delta + A) > 1$.

Additional remarks

Logarithmic utility. With $\sigma = 1$, utility is $\log(c)$. Let $R = 1 - \delta + A$ and guess $v(k) = b_0 + b_1 \log(k)$. The Bellman equation is

$$b_0 + b_1 \log(k) = \max_{k'} \left\{ \log(Rk - k') + \beta \left(b_0 + b_1 \log(k') \right) \right\},$$

and the first-order and envelope conditions are

$$\frac{1}{Rk - k'} = \frac{\beta b_1}{k'},$$
$$\frac{b_1}{k} = \frac{R}{Rk - k'}.$$

Using the first condition to replace $Rk - k'$ in the second gives the policy function $k' = \beta R k$, so that $c = (1 - \beta) R k$ and $b_1 = 1/(1 - \beta)$. Plugging the policy function into the Bellman equation and matching the constants gives

$$b_0 = \frac{\log R + (1 - \beta) \log(1 - \beta) + \beta \log \beta}{(1 - \beta)^2}.$$

This is the continuation of the CRRA solution at $\sigma = 1$: the policy function $k' = (\beta R)^{1/\sigma} k$ converges to $\beta R k$, and $a = \left(R^{(\sigma-1)/\sigma} - \beta^{1/\sigma} \right)^{-\sigma}$ converges to $1/(1 - \beta) = b_1$. The value functions match only after subtracting from the CRRA one the constant $1/((1 - \sigma)(1 - \beta))$, which does not affect choices: as $\sigma \to 1$, $(a k^{1-\sigma} - 1/(1 - \beta))/(1 - \sigma)$ converges to $b_0 + \log(k)/(1 - \beta)$. With logarithmic utility the saving rate β does not depend on the return R, because the income and substitution effects of the return cancel, and the restriction $\beta R^{1-\sigma} < 1$ becomes $\beta < 1$, which always holds. The two cases compare as follows:

	CRRA, $\sigma \neq 1$	log, $\sigma = 1$
Growth factor of k and c	$(\beta R)^{1/\sigma}$	βR
Saving rate $k'/(Rk)$	$\beta^{1/\sigma} R^{(1-\sigma)/\sigma}$	β
Value function	$a k^{1-\sigma}/(1 - \sigma)$	$b_0 + \log(k)/(1 - \beta)$
Restriction	$\beta R^{1-\sigma} < 1$	$\beta < 1$

Under CRRA the value at $k = 1$ enters multiplicatively, $v(k) = v(1)k^{1-\sigma}$, because $u(\lambda c) = \lambda^{1-\sigma}u(c)$; under logarithmic utility it enters additively, $v(k) = v(1) + \log(k)/(1-\beta)$, because $\log(\lambda c) = \log(\lambda) + \log(c)$, so that $b_0 = v(1)$. With the normalised utility $(c^{1-\sigma} - 1)/(1-\sigma)$, the CRRA value function would also have an additive constant, $-1/((1-\sigma)(1-\beta))$, which diverges as $\sigma \to 1$.

The story of this model. The *AK* model is due to Rebelo (1991). It was developed to study how policy affects long-run growth, a question the neoclassical growth model of Question 2 cannot address: there, the long-run growth rate is pinned down by the exogenous rate of technological progress, $\gamma^{1/(1-\alpha)}$, and policy can only shift the level of the balanced growth path. In the *AK* model, capital is interpreted broadly (physical and human capital), so returns to capital do not diminish, and the growth rate depends on the return to capital, which policy can affect. For example, a tax τ on capital income, with the revenue rebated lump-sum, lowers the after-tax return to $1 - \delta + (1-\tau)A$ and hence the growth rate to

$$(\beta(1 - \delta + (1-\tau)A))^{\frac{1}{\sigma}} - 1.$$

Higher capital income taxes therefore lower growth permanently, not just temporarily.

Transitional dynamics. Unlike the neoclassical growth model, the *AK* model has no transitional dynamics: whatever the initial capital stock k_0, capital, output, and consumption grow at the constant rate $(\beta(1 - \delta + A))^{1/\sigma} - 1$ from period 0 onward. As a result, there is no convergence: two economies with the same parameters but different k_0 grow at the same rate, and their initial income gap persists forever. Similarly, an earthquake that destroys part of the capital stock permanently lowers the level of the growth path, without any catching up (compare with Figure 1).

A question to think about. The Euler equation pins down the growth rate of consumption, $c_{t+1}/c_t = (\beta(1 - \delta + A))^{1/\sigma}$, but not its initial level c_0. Which value of c_0 is optimal, and what goes wrong with the others? *Hint:* given $c_t = c_0(\beta(1 - \delta + A))^{t/\sigma}$, solve the resource constraint forward for k_t, and use the transversality condition and the requirement that $k_t \geq 0$. Check that your answer is consistent with the policy function found in part 2.

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